

Assignment 1: E-Purchase Business Process Modeling

BPMN 2.0 and Petri-net models for an Amazon/Souq-like order process

Submission focus

This report models the same business process in two accepted ways: BPMN 2.0 for business readability and Petri nets for formal token-based behavior. The BPMN evidence includes an explicit collaboration with separate participant pools, internal e-commerce lanes, data objects, data stores, data associations, and message flows between organizations. It includes the as-is process, a to-be improvement, issue analysis, and exported evidence files.

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1 Executive Scope

The modeled process covers an e-purchase from order submission through validation, stock handling, payment, fulfillment, delivery, complaint handling, and final archiving. To maximize assignment coverage, the same process is documented in both required modeling styles:

- BPMN 2.0 models for the as-is and to-be business process.
- Petri-net models for the same as-is and to-be behavior, including choices, synchronization, loops, and final markings.

The to-be design improves early validation, parallelizes independent checks, connects complaints to the same order case, and adds proactive SLA monitoring for delayed goods.

2 Prepared Evidence Assets

File	Purpose
output/assignment1_epurchase_as_is.bpmn	BPMN 2.0 source for the as-is model, including separate participant pools, internal lanes, message flows, data objects, data stores, and associations, importable into bpmn.io or Camunda.
output/assignment1_epurchase_to_be.bpmn	BPMN 2.0 source for the improved to-be model with the same complete collaboration and notation coverage.
output/assignment1_epurchase_as_is_petri.dot and .pnml	Graphviz and PNML versions of the formal as-is Petri net.
output/assignment1_epurchase_to_be_petri.dot and .pnml	Graphviz and PNML versions of the improved to-be Petri net.
screenshots/assignment1/*.svg	Browser/tool exports used as the visual evidence in this report.

3 Evaluation Criteria Coverage

The assignment rubric is used as an explicit checklist. The report covers each marking category as follows:

Criteria category	Marks	How this report covers it
Process model quality	20	BPMN 2.0 models use separate collaboration pools for Customer, E-Commerce Operations, Payment Service, Warehouse/Fulfillment, and Courier/Shipment; internal e-commerce lanes; start/end events; tasks; gateways; timer/message intermediate events; data objects; data stores; data associations; internal sequence flows; and straight dashed cross-pool message flows. Petri nets also model the process with places, transitions, arcs, initial marking, final markings, choices, joins, and loops. Core e-purchase steps are shown: order capture, stock check, address validation, payment authorization, invoice, payment receipt, packing, shipping, delivery, complaint, and archive.

Tool proficiency	5	BPMN source files include collaboration, participant pool, lane, message-flow, data object, data store, and association metadata and are prepared for bpmn.io/Camunda import. Petri-net DOT/PNML files are prepared for Graphviz, diagrams.net, or PNML tools. The submitted report includes SVG exports rendered into the PDF through the verified cache.
Use of intermediate events	10	The as-is BPMN uses a delivery-confirmation intermediate message event and a delayed-goods timer event. The to-be BPMN uses the same material-covered event types for SLA breach handling. Both models route delayed goods into complaint registration, customer notification/escalation, and return to delivery tracking.
Documentation and clarity	15	Every diagram is labeled, introduced, and followed by text explaining what the model means. Participants, responsibilities, formal Petri-net summary, file evidence, and case coverage are documented.
Analysis before enhancement	10	The as-is issue register identifies reactive complaints, late address correction, weak synchronization, separated complaint ownership, and late archiving. Each issue includes consequence, type, and justification through its proposed fix.
Enhancement suggestions	10	The to-be model moves validation earlier, parallelizes stock/payment checks, standardizes fulfillment, adds tracking, connects complaints to the order case, and makes delay handling proactive.
Comparative analysis	10	The before/after comparison explains what changed and why. Expected outcomes are added for shorter waits, lower rework, clearer customer communication, and stronger auditability.
Report quality and submission	10	The report has a cover page, contents, list of figures, consistent table styling, labeled diagrams, and no LaTeX-drawn diagrams. The repository is cleaned after compilation.
Originality and effort	10	The process is a realistic Amazon/Souq-like scenario with payment service, warehouse, courier, support, complaint escalation, SLA tracking, audit events, and two modeling methods rather than only the minimum model.

4 Modeling Standard and Core Step Checklist

Requirement	Implementation evidence
Correct BPMN start/end behavior	The as-is model starts at <i>Order received</i> and reaches either <i>Order rejected</i> or <i>Order closed</i> . The to-be model starts at <i>Checkout submitted</i> and has the same accepted/rejected closure logic.
Core e-purchase flow	Both versions include order validation, stock decision, payment authorization, invoice emission, payment receipt, warehouse fulfillment, shipment, delivery confirmation, and archiving.
Decision logic	Exclusive gateways model stock availability, address/order validity, delivery/SLA status, and rejection paths. Lower alternatives leave the gateway from the bottom, while the main/positive path leaves from the right. Parallel gateways model only true independent split/join behavior.
Goods-delay event	The delayed-goods path is not only mentioned in text; it is modeled as an event-driven complaint/escalation branch.
Petri-net formal semantics	The Petri nets include initial/final markings, routing transitions, parallel split/join behavior, and a loop from delay escalation back to delivery waiting/tracking.
Online-tool evidence	SVG exports are stored in <code>screenshots/assignment1/</code> ; importable BPMN, DOT, and PNML source files remain in <code>output/</code> . The BPMN SVGs visibly draw separate pool boundaries with gaps, internal lane headers, straight dashed cross-pool message flows, data artifacts, and dotted data associations.

5 Lecture Concepts Applied

The report follows the lecture definitions of a business process as structured work involving people, systems, data, and resources to deliver value to a customer. The BPMN lecture is used for events, tasks, gateways, sequence flows, and process readability. The Petri-net lecture is used for places, transitions, arcs, markings, reachability, synchronization, and loops. The process-analysis lecture is used for issue registers, customer interfaces, quality risks, cycle-time reduction, and improvement opportunities.

6 BPMN 2.0 Model

6.1 BPMN Pools, Lanes, and Data Artifacts

The BPMN 2.0 source files are collaborations, not one large pool. Customer, E-Commerce Operations, Payment Service, Warehouse/Fulfillment, and Courier/Shipment are modeled as separate participant pools. Only the E-Commerce Operations pool is subdivided into internal lanes: Marketplace system and Customer support. Handoffs between different pools are shown as dashed BPMN message flows, while solid sequence flows stay inside the owning participant.

Notation element	Where it appears in the models
Pools	Separate pools represent Customer, E-Commerce Operations, Payment Service, Warehouse/Fulfillment, and Courier/Shipment.
Lanes	Marketplace system and Customer support are internal lanes inside the E-Commerce Operations pool. External organizations are separate pools, not lanes.
Message flows	Straight dashed cross-pool flows model order submission, early stock/payment checks, payment handoff, warehouse request, shipment handoff, and completion status messages.
Data objects	Order request, customer address, payment record, invoice, pick list, checkout/order record, customer profile, shipping label, and complaint case appear beside the tasks that read or update them.
Data stores	Product/stock or inventory records, shipment/tracking records, and order archive/audit-log stores show persistent system data.
Data associations	Dotted BPMN associations connect tasks and gateways to the same-pool data artifacts they consume, create, or update.

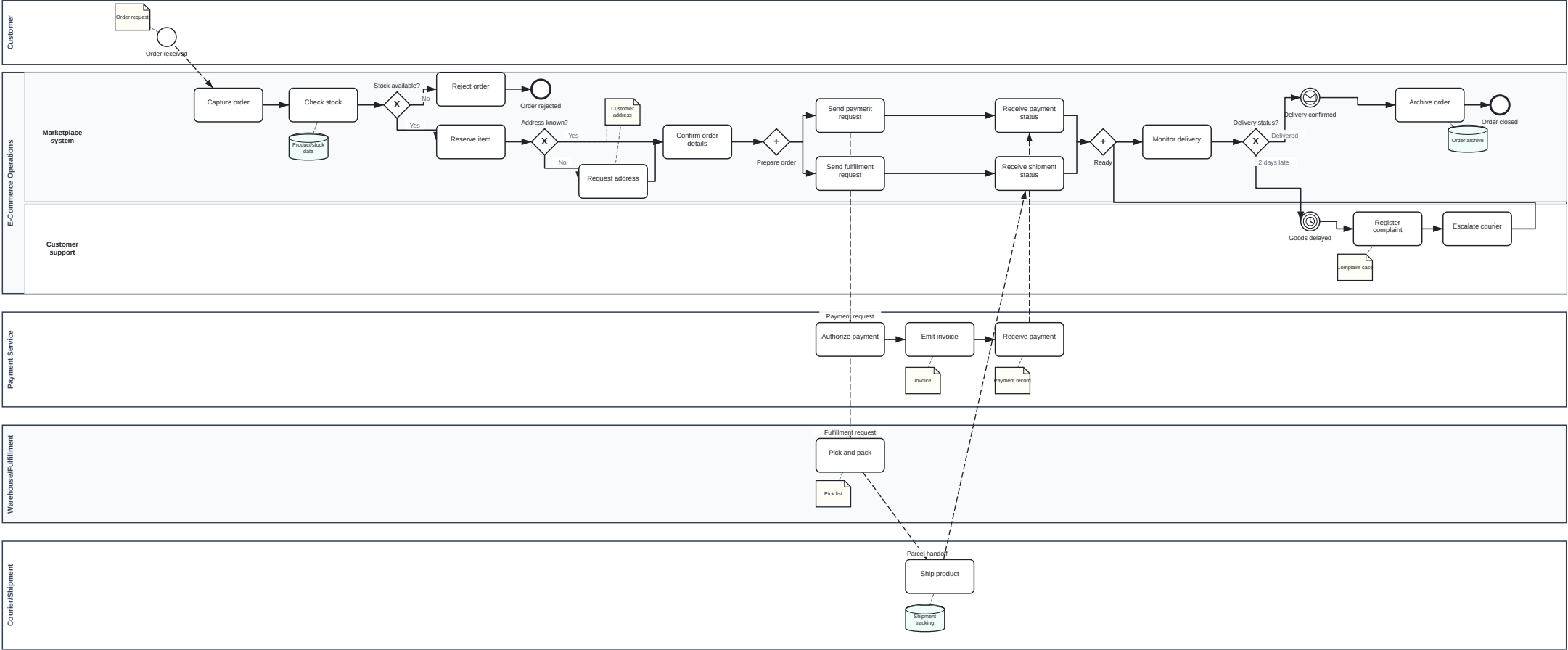
6.2 Participants and Responsibilities

Participant or lane	Responsibility
Customer pool	Submits checkout/order data and receives customer-facing delivery/complaint communication.
E-Commerce Operations pool	Owns order capture, validation, inventory decisions, tracking, archiving, SLA monitoring, and complaint coordination.
Marketplace system lane	Internal e-commerce lane for system checks, status tracking, gateways, and archive/audit actions.
Customer support lane	Internal e-commerce lane for complaint registration, customer notification, and courier escalation.
Payment Service pool	Authorizes payment, emits invoice/payment evidence, and returns payment status to e-commerce.
Warehouse/Fulfillment pool	Reserves fulfillment capacity, creates pick/pack evidence, and hands the parcel to shipment.
Courier/Shipment pool	Ships the product and updates shipment/tracking evidence used by e-commerce monitoring.

6.3 As-Is BPMN

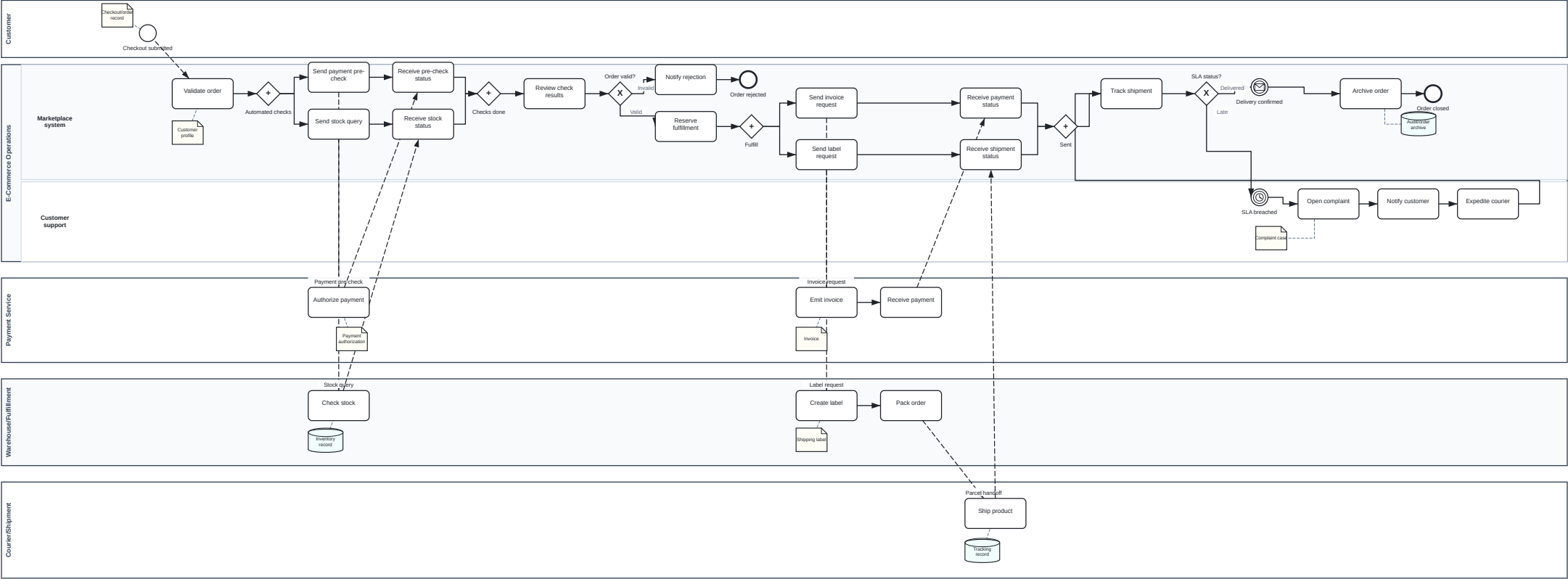
The as-is process starts when the order is received. The system captures the order and checks stock. If stock is unavailable, the order is rejected. If stock is available, the item is reserved, address data is checked, and missing address data is requested. The two address outcomes meet at a confirmation task before the e-commerce pool splits into parallel handoff branches. Each branch uses explicit BPMN send/receive tasks: payment request then payment status, and fulfillment request then shipment status. Dashed message flows connect those handoff tasks to the external Payment Service, Warehouse/Fulfillment, and Courier/Shipment pools. After both statuses synchronize, the process monitors delivery status.

The delayed-goods scenario is represented by an intermediate timer event after shipment. If delivery confirmation is not received within the SLA, the timer branch registers a complaint, escalates the courier, and loops back to delivery monitoring before the next delivery-status decision. The delivery confirmation event remains a single catch event owned by the delivery wait.



6.4 To-Be BPMN

The to-be process moves validation earlier, runs stock and payment checks in parallel, reviews the combined check results, standardizes fulfillment preparation, adds explicit payment receipt, adds shipment tracking, and makes the complaint path proactive. The early check section uses explicit e-commerce send/receive tasks for payment pre-check and stock status, which fill the marketplace lane between the automated-check split and the checks-done join. The fulfillment section uses explicit e-commerce send/receive handoff tasks for invoice/payment status and label/shipment status, so the collaboration shows both internal sequence flow and straight cross-pool message flow. The SLA breach event opens a complaint automatically, notifies the customer, expedites the courier, and then loops back to shipment tracking before the next SLA-status decision.



7 Petri-Net Model

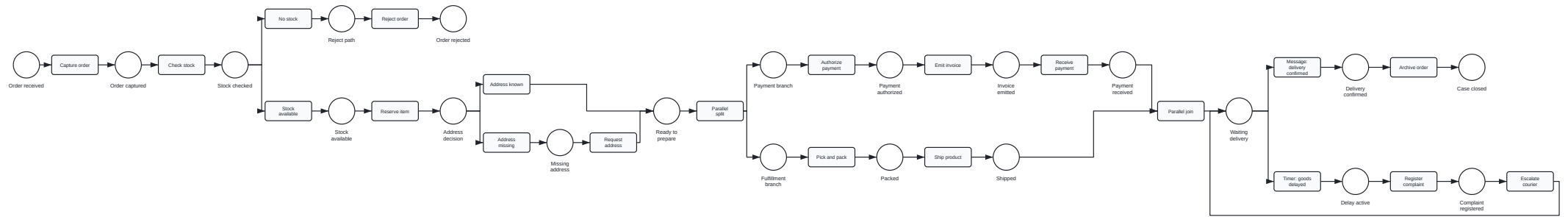
The Petri-net model is the second modeling method. States are represented as places, activities and routing decisions as transitions, and allowed movement as directed arcs. One token starts in the initial place. Routing choices, parallel splits, and joins are shown as normal transitions using the same visual style as the other transitions.

7.1 Formal Net Summary

Model	Places	Transitions	Arcs	Initial and final markings
As-is	21	21	44	$M_0 = \{p_start\}$; accepted final $\{p_done\}$ or rejected final $\{p_rejected\}$
To-be	25	23	50	$M_0 = \{p_start\}$; accepted final $\{p_done\}$ or rejected final $\{p_rejected\}$

7.2 As-Is Petri Net

The as-is Petri net has one token in **p_start**. The stock decision place enables either rejection or reservation. Rejection ends at the visible **p_rejected** place. Both address outcomes produce the same preparation place, matching the corrected BPMN confirmation step. The preparation split produces one token for the payment branch and one token for the fulfillment branch; the join requires payment receipt and shipment before delivery waiting can continue. The goods-delay timer transition creates the complaint token, and the escalation transition loops back to the delivery waiting place.

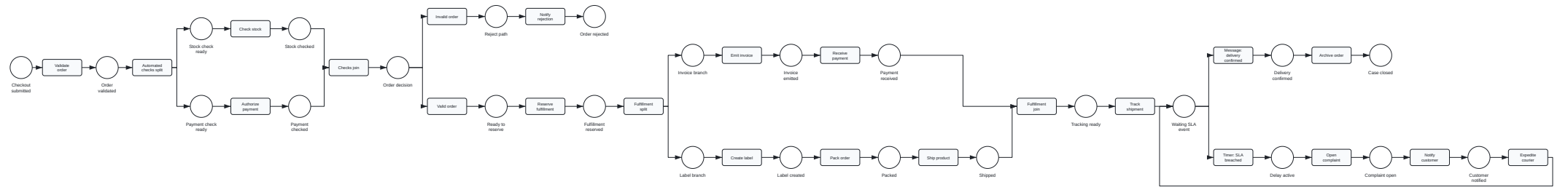


Assignment 1 As-Is Petri Net

7.3 To-Be Petri Net

The to-be Petri net shows two improvements formally. First, stock check and payment authorization are enabled together after validation, then synchronized before the order validity decision; rejected orders end at **p_rejected**. Second, invoice/payment receipt and physical fulfillment are split into parallel branches and joined before tracking. The SLA breach transition sends the case through complaint opening, customer notification, and courier expedition before returning to the SLA wait place.

Assignment 1 To-Be Petri Net



8 As-Is Analysis

No.	Issue	Consequence	Type	Proposed fix
1	Complaint is reactive	Customer may discover the delay before the company reacts.	Customer interface	Add proactive shipment tracking and SLA monitoring.
2	Address correction is late	Reserved stock may wait while customer data is fixed.	Data/process	Validate address before reservation.
3	Weak synchronization	Invoice or shipment may wait for manual status checking.	IT/process	Automate parallel checks and clear joins.
4	Complaint path is separate	Support may reconstruct context manually.	Organization	Attach complaint to the same order case and assign ownership.
5	Archiving only at closure	Monitoring data is late and hard to mine.	IT	Record status events throughout execution.

9 Enhancements and Comparative Analysis

Area	As-is process	To-be improvement
Checkout data	Address may be requested after stock reservation.	Validate address and order data before stock is held.
Cycle time	Several checks can wait sequentially.	Run stock and payment checks in parallel.
Delay handling	Complaint begins after a late-delivery event.	SLA monitoring opens the complaint and notification path proactively.
Customer quality	Customer may need to ask for status.	Tracking events and notifications are part of the process.
Cost and rework	Support may manually reconstruct order context.	Complaint is connected to the same order case.
Modeling rigor	BPMN shows business readability.	Petri nets add formal token semantics for choices, joins, and loops.

9.1 Expected Outcomes from the To-Be Model

Improvement target	Change made	Expected outcome
Shorter waiting before fulfillment	Validate order data first and run stock/payment checks in parallel through explicit send/receive status tasks.	Fewer orders reserve inventory while still missing data; independent checks no longer wait unnecessarily.
Faster delay response	Add SLA monitoring and proactive complaint opening.	The company reacts before the customer has to chase the order, improving customer experience.
Lower manual rework	Keep complaint handling attached to the same order case.	Support sees order, shipment, and complaint context together instead of reconstructing the case manually.
Better audit trail	Record status events before final archiving.	Later process mining can identify bottlenecks, courier delays, and repeated failure points.

10 Case Coverage

- Normal order: checkout, validation, stock/payment checks, fulfillment, delivery confirmation, and archive.
- Rejected order: invalid payment, invalid data, or unavailable stock leads to rejection.
- Missing address: handled before stock is blocked in the to-be model.
- On-time delivery: delivery message event continues to archive.
- Goods delayed: timer/SLA event opens complaint, notifies customer, escalates courier, and loops back to delivery tracking.
- Audit trail: status updates and archiving support later monitoring and process mining.

11 Conclusion

The BPMN models provide the business-readable view required for process discussion, while the Petri nets provide the formal execution view required for reasoning about choices, concurrency, synchronization, and loops. The to-be model directly addresses the main weaknesses in the as-is process: late validation, reactive complaint handling, and weak status visibility.